

BUSINESS INTELLIGENCE AND TOOLS

EXPERIMENT - 2

Problem File

Title: Designing and Implementing ETL Processes Using Microsoft Power BI

Aim: To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

Objectives

- To understand the ETL process.
- To extract data from multiple worksheets.
- To perform data transformation using Power Query Editor.
- To create relationships among tables.
- To develop analytical measures using DAX.
- To prepare a dataset for reporting and decision-making.

Tools Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

Theory:

ETL (Extract, Transform, Load): ETL is a fundamental process in Business Intelligence that allows organizations to collect data from different sources, transform it into a clean and consistent format, and load it into a centralized environment for analysis.

Extract Phase: The extraction phase involves collecting data from source systems. In this experiment, the Orders, Returns, and People worksheets are imported into Power BI.

Transform Phase: The transformation phase includes:

- Data cleaning
- Duplicate removal
- Handling missing values
- Data type correction
- Creating calculated columns
- Data standardization

Load Phase: The transformed data is loaded into the Power BI data model, where relationships and DAX measures are created.

Expected Outcome

A clean, transformed, and integrated dataset that can be used for reporting and business analysis.